SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I P.Sunayana(192424303) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

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1)

Security Attacks, Threats, Risks and Security Goals (concept Map)

Security Attacks

Passive Attacks
- Eavesdropping.
- Traffic analysis.
- Monitoring.

Active Attacks
- Modification.
- Replay.
- Masquarade.
- Denial of service.

Threats
- Malware (viruses, worms)
- Phishing / social engineering.
- Inside threats
- Natual disasters.
- Human error, misconfiguration.

Risks
- Data loss / data breach.
- Financial loss.
- Reputational damage.
- Service disruption.
- Legal / compliance issues.

INFORMATION SECURITY

Relationships
- Threats exploit vulnerabilities
- Attacks cause harm.
- Risks represent potential losses.
- Security goals help reduce risks.

Security Goals (CIA Triad)

Confidentiality (prevents unauthorized access)

Integrity (prevents unauthorized modification)

Availability (ensures services are accessible).

Real-world example

- A phishing email steals online bank banking credentials, leading to unautho-rized access & possible financial loss.

Mitigation:- Multi-factor authentication, encryption.

2)

Substitution Techniques (Concept Map)

SUBSTITUTION TECHNIQUES.

Caesar cipher
- Shift each letter by a fixed number.
- key: shift value (e.g., 3)
- Eg:- HELLO → KHOOR
- Weak against brute - force attacks.

Monoalphabetic cipher
- Replaces each letter using a fixed substitution alphabet.
- key:- 26 letters.
- Eg:- A → Q, B → W, ...
- Vulnerable to frequency.

Vigenere cipher
- uses a repeating keyword to shift letters by different amounts.
- key :- KEY.
- Eg :- HELLO → RIJVS.
- Stronger than simple substitution.

Hill cipher.
- Uses a matrix multiplication & arithmetic.
- key :- key matrix.
- Eg:- Encrypts blocks of letters.
- Stronger transformation.

Comparsion :-

Key usage	Shift value	Substitution alphabet	keyword	key matrix
Complexity	low	Moderate	Moderate	Higher
Security strength	weak	weak	Stronger	Stronger.

Key Takeaway :-
- All are substitution techniques but differ in key usage and complexity.
- Hill & vigenere ciphers are generally stronger than caesar, but none are secure for modern sensitive communication.

3)

Classical Encryption Schemes (Concept Map)

Classical Encryption Schemes

Substitution Techniques
(Replace letters / symbols, keep position)

Examples:-
- Caesar cipher.
- Monoalphabetic cipher.
- Vigenere cipher.
- Hill cipher.

Transposition Techniques

Examples:-
- Rail Fence cipher.
- Columnar Transposition cipher.

Differences
- Substitution changes the symbols, transposition changes the order.
- Substitution may preserve some letter-frequency patterns.
- Transposition preserves the original letter frequencies.

Similarities
- Both uses key & encryption/decryption processes.
- Both aim to provide confidentiality.
- Both are vulnerable to crypt-analysis when used alone.

Real-world Illustration.
- Caesar cipher:-
  HELLO → KHOOR
  (shift = 3).
- Transposition cipher:-
  HELLO → HLOEL
  (rearranged order).

4)

Steganography Concepts
(Concept map)

Steganography.

Hiding information in ordinary files.

Data hiding
- Conceals secret information inside another file / medium.
- Techniques:- LSB, Spatial-domain, transform-domain.
- Example:- Hiding a text message in an image / audio.

Watermarking
- Embeds identifying information into digital content.
- Types:- Visible, Invisible, robust, fragile.
- Example:- Adding a digital watermark to a photograph.

Survivability
- Measures whether hidden information remains detectable after processing/modification.
- Factors:- Robustness, capacity, security.
- Example:- Robust watermark.

Relationships :-
- Data hiding conceals a message.
- Watermarking embeds identifying information.
- Survivability determines how well the data changes.

Conclusion:- Steganography supports secure communication by hiding information within the ordinary digital files, while watermarking & Survivability helps to protect & verification.